Networking Subsystem Parameters

Here are a set of networking parameters (adopted from source code), which also mention those that are tunable at compile time. Some of the terms used in the descriptions are as follows:

- RIF cache: It is used while managing the source route information.
- mrouted: It meddles with the multicast router (pronounced 'm route d')—essentially a daemon that does the forwarding of IP multicast datagrams.
- vif: Virtual interface
- iovecs: These are used if the data has to be written atomically (Atomic Data Access will be discussed in later columns).
- PE1CHL: Information about PE1CHL can be found at Rob's website, http://pe1chl.xs4all.nl/

Item	Description
MAX_LINKS	Maximum number of netlink minor devices (1-32)
RIF_TABLE_SIZE	Token-ring RIF cache size (tunable)
AARP_HASH_SIZE	Size of AppleTalk hash table (tunable)
AX25_DEF_T1 AX25_DEF_T2 AX25_DEF_T3 AX25_DEF_N2	AX.25 parameters. These are all tunable via SIOCAX25SETPARMS. T1-T3, N2 have the meanings in the specification
AX25_DEF_AXDEFMODE	8 = normal; 128 is PE1CHL extended
AX25_DEF_IPDEFMODE	'D' - datagram 'V' - virtual connection
AX25_DEF_BACKOFF	'E'xponential/'L'inear
AX25_DEF_NETROM	Allow netrom; 1=Y
AX25_DF_TEXT	Allow PID=Text; 1=Y
AX25_DEF_WINDOW	Window for normal mode
AX25_DEF_EWINDOW	Window for PE1CHL mode
AX25_DEF_DIGI	1 for in-band; 2 for cross-band; 3 for both
AX25_DEF_CONMODE	Allow connected modes; 1=Yes
AX25_ROUTE_MAX	AX.25 route cache size; not currently tunable
Unnamed (16)	Number of protocol hash slots (tunable)
DEV_NUMBUFFS	Number of priority levels (not easily tunable)
Unnamed (300)	Maximum packet backlog queue (tunable)
MAX_IOVEC	Maximum number of iovecs in a message (tunable)
MIN_WINDOW	Offered minimum window (tunable)
MAX_WINDOW	Offered maximum window (tunable)
MAX_HEADER	Largest physical header (tunable)
MAX_ADDR_LEN	Largest physical address (tunable)
SOCK_ARRAY_SIZE	IP socket array hash size (tunable)
IP_MAX_MEMBERSHIPS	Largest number of groups per socket (BSD style) (tunable)
16	Hard-coded constant for amount of room allowed for cache align and faster forwarding (tunable)
IP_FRAG_TIME	Time we hold a fragment for (tunable)
PORT_MASQ_BEGIN	First port reserved for masquerade (tunable)
PORT_MASQ_END	Last port used for masquerade (tunable)
MASQUERADE_EXPIRE_TCP_FIN	Time we keep a masquerade for after a FIN
MASQUERADE_EXPIRE_UDP	Time we keep a UDP masquerade for (tunable)
MAXVIFS	Maximum mrouted vifs (1-32)
MFC_LINES	Lines in the multicast router cache (tunable)

the function returns *EINVAL* or *EFAULT* if the buffer size is negative, or if it is too large or inaccessible. You may note that the function copies the address to a given limit specified by the user and the actual length the data is written over the specified length limit. If the allocation by the user is incorrect, it will cause corruption.